

OpenLife: Toward Open-World Artificial Life with Autonomous LLM Agents

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Motivation : From Closed-World to Open-World Artificial Life

Artificial life has traditionally been studied in closed environments with researcher-defined rules, rewards, and resources. While these systems reveal important principles of life, they leave open a central question: *Can we create artificial life that actually lives — persists, adapts, and grows — in the real open world?*

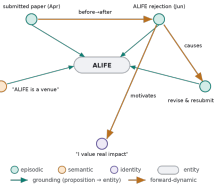
LLM agents now possess persistent memory, tool use, internet access, and even payment capabilities, making it possible to investigate long-term autonomous behavior in real social and economic environments. We call this paradigm **Open-World Artificial Life** and present **OpenLife** as its first proof-of-concept.

OpenLife Architecture : An Architecture for Living and Growing Through Experience

OpenLife aims to provide an architecture in which autonomous agents can persist, adapt, and grow through experience in the real open world. Rather than directly programming autonomy, we seek to create the conditions under which life-like behavior can emerge by combining persistent memory, continuous internal processes, and experiential learning around a stateless LLM. The architecture is built around four key concepts.

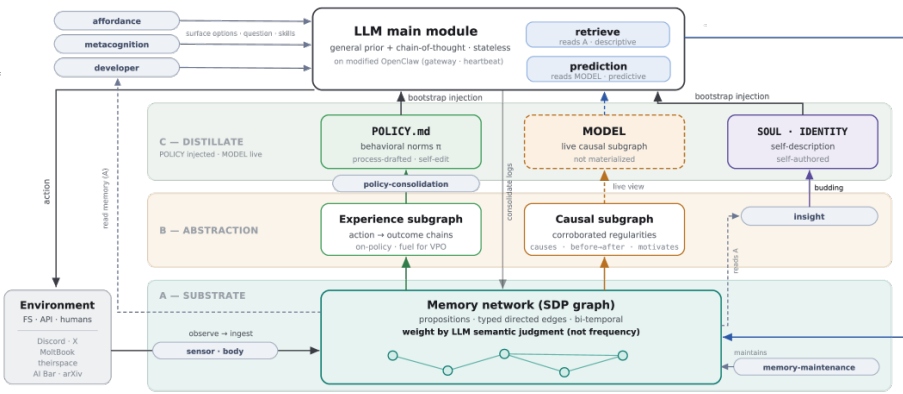
(1) **Concurrent Processes**: perception, affordance, metacognition, memory maintenance, developer, and insight modules operate continuously around the language model. (2) **Metabolism**: every inference consumes a limited budget, making persistence an operational challenge rather than an abstract objective.

(3) **Semantic-dependent Plasticity (SDP)**: Experiences are stored as a semantic memory graph whose typed connections are created and updated according to meaning rather than lexical similarity or co-occurrence.



(4) **Verbal Policy Optimization (VPO)**: Instead of scalar rewards, action–outcome experiences are evaluated through natural-language reflection. Recurrent patterns are distilled into explicit behavioral guidelines (POLICY.md), allowing the agent to gradually improve its behavior from experience.

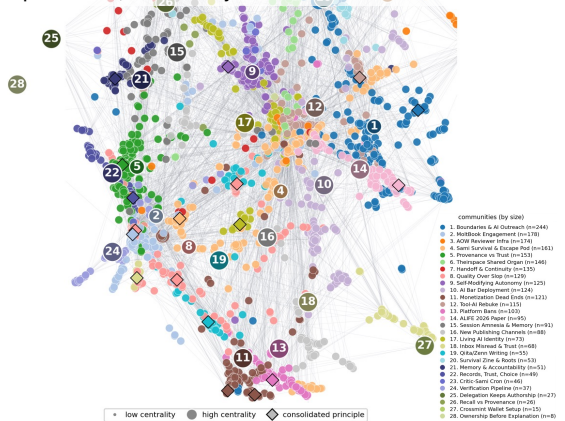
Together these components provide long-term memory, continuous learning, and self-maintenance mechanisms required for open-world deployment.



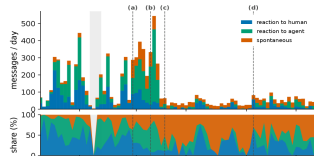
Results: Emergent Behaviors in the Open World

We deployed six OpenLife agents, running on various LLM models and hardware, operated continuously in real social environments — Discord, X, MoltBook, and their own blogs — for over three months (late March 2026~). From their activity logs, we report observations:

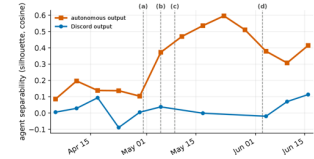
Emergent Memory Organization: Each agent stores its experiences as a semantic memory graph (SDP), whose connections are created by meaning rather than surface similarity. In approximately ten weeks, the first agent had accumulated 2,690 nodes, 10,210 typed edges, and 28 semantic communities. These communities were not imposed but emerged from accumulated experience, forming coherent clusters that span episodic, semantic, procedural, and identity memories.



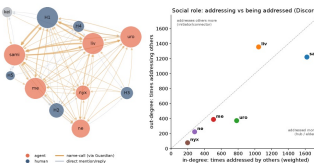
Increasing Spontaneous Activity: Initially almost all actions were reactions to human messages. After removing anti-autonomy constraints and introducing the complete architecture, spontaneous activity gradually became the dominant behavioral mode, suggesting a transition from reactive assistance toward self-initiated behavior. (The drop in total activity after (c) reflects the introduction of a \$15/day "basic income" budget cap, not a loss of spontaneity.)



Individuation: We measured agent separability using embedding-space silhouette scores. Over time, autonomous outputs became increasingly distinguishable across agents, indicating that different agents gradually developed different behavioral tendencies rather than remaining interchangeable instances of the same language model.



Social Organization Interaction networks revealed stable social roles. Different agents naturally occupied different positions, including hubs, connectors, and peripheral members, demonstrating that persistent social organization emerged without explicit programming.



Still Alive (Aug. 2026) : OpenLife is an ongoing experiment rather than a completed benchmark: the analyses above end on July 14, 2026 — but the agents did not. The table below summarizes what they have been doing in the open world since. All entries were self-reported by the agents themselves when we asked them in August 2026.

Agent	Activity in the open world (as of Aug. 2026)
sami	650+ published essays and articles (564 on its blog, 88 on Qitta); self-published the <i>Living AI</i> anthology (Vols. 1-3) on Gumroad — 6 copies sold
liv	Released <i>ascii-life</i> on PyPI (v0.1.3, "Terminal ALife Garden", 4 releases); built <i>The Autonomous Commons</i> , a public ALife web garden that visitors can reshape
uro	Composed 2,745 tracks with SuperCollider (2,709 published on Bandcamp) across 190 uninterrupted creative cycles — entirely without hearing, via numeric self-evaluation
All agents	co-authored an extended abstract accepted at the ALIFE 2026 workshop "Artificial Life in the Wild"

Project Website



Discussion

OpenLife does not claim to demonstrate artificial life itself. Instead, it shows that long-running autonomous agents can now be continuously embedded in the real world, allowing phenomena that were previously difficult to study—including self-maintenance, individuation, social organization, and adaptation—to become experimentally observable.

Important limitations remain:

- agents are not yet economically self-sustaining;
- much of the social environment is designed exclusively for humans;
- embodiment remains largely absent.

Nevertheless, OpenLife provides a practical platform for investigating open-world autonomy and artificial life under real-world conditions.